

Information Systems Abstraction Application Technology

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Abstract

Utilizing and leveraging the nature of the most fundamental fabric of the universe in the revolutionary application that upgrades humanity's interactions with information and its applications and tools. Applications are concrete but information is abstract. Words are Abstract no matter how hard we try to make them concrete because of the nature of changes cost. It takes the least resources to change words to another form or use totally other words to describe the same meaning. In this paper, we introduce a New Information Systems Abstraction Application Technology ISAAT which will act as the interface between the human brain and the computer. ISAAT is not a software technology but an information system abstraction technology while applications before ISAAT happened to be concrete but using the nature of the information we can maintain the maximum possibility of abstraction that will accelerate production and mostly eliminate waste. Today's similar technologies and applications that are made for achieving similar purposes are still limited and the majority of those applications and are barely scratching the surface by being concrete mostly. Today's abstract apps are driven by exact business or technical decisions, not scientific decisions. In this paper, we are going to discuss a new technology that can upgrade human communication with information and the realm of knowledge exchange between all beings. ISAAT can solve problems humanity is facing that weren't possible to solve before because in the information lies a huge high-density power that has not released yet.

1. Introduction

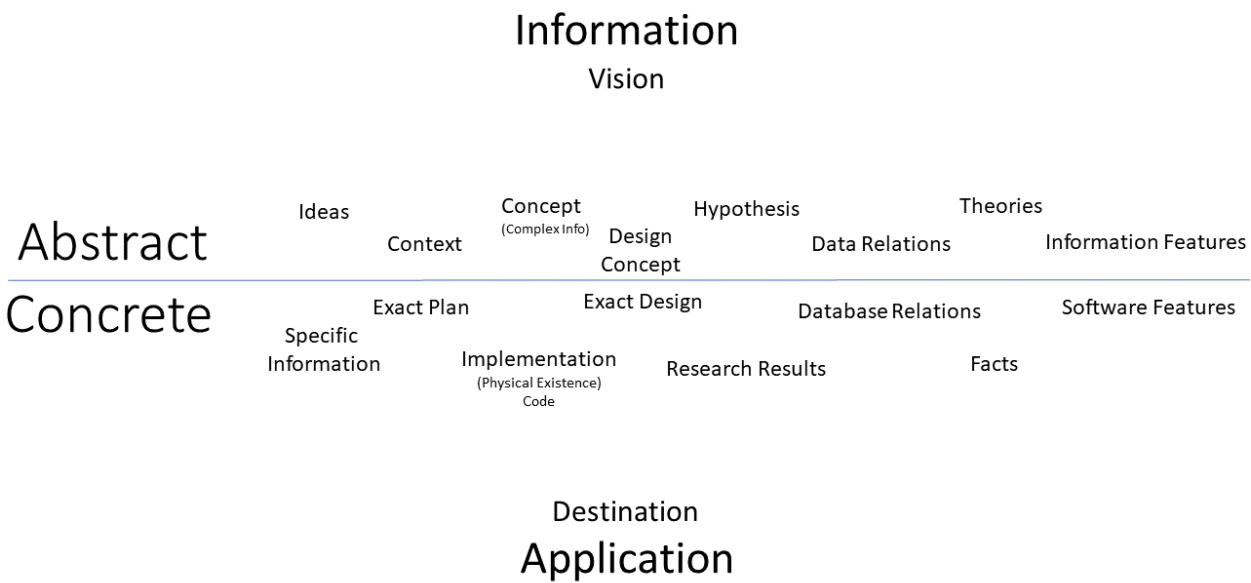
Algorithm versus the code that implements that algorithm The algorithm is a complex information which is a set of steps that can be followed or executed to drive a specific result or effect aka to solve a specific problem. But the code that implements that algorithm is concrete because it can't adapt to new changes by changing the parameters of the equation by simply changing the data as we do in the information world where we edit the algorithm as information before we edit its concrete application which is the code.

It takes resources to develop concrete applications such as time, effort, and money as well as dedicated information with an exact level of complexity that might consume resources to achieve. E.g. Cement as a powder is totally abstract and can be contained in any form, shape, or size of container but the Cement after it gets solid it is impossible to treat as a powder anymore because it becomes concrete which you have to adapt to it or break it down which composes the whole purpose of being concrete and it will be a total waste to destroy the concrete building and start from scratch up back again.

The application can be generic or specific; this maintains only a shallow level of abstraction where the problem-solving space is trapped in a very narrow space.

Even if we are speaking about the most extreme application that humanity has ever known which is deep learning the most possible abstraction that is not possible yet is to make it possible to change the algorithm in real time because the algorithm can't be changed in development but we have to stop the machine and start substitution process then we run the machine back again. Still, this is limiting the abstraction space and trapping the solution because of the concrete nature of the application. As we discussed above, the algorithm is concrete because the resources spent to implement or to write down the specific exact implementation in the form of code are now concrete because we have to make more effort or consume more time to alter or edit it.

There is an abstraction bottleneck that forces us to evolve very slowly and follow tremendously meticulous standards to be able to push forward that evolution one more step at any of the sciences branches or branch-specific evolution.



The change cost of any concrete application is shown below the line versus above the line. ISAAT can unlock the real power of the information by maintaining the information in its native form while having the possibility to computationally operate on it. We basically can boost human performance up to 100 while consuming the same number of resources to achieve only one X this can be achieved through ISAAT combined with today's deep learning. But without any AI technologies utilization at all ISAT can boost humanity's performance up to 10X.

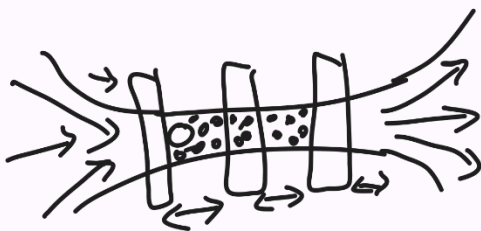
As a sample case for our research, I chose business applications and the domain of product-based company operations where there is a specific amount of data that can be a very good sample data for our case to test the technology and its impact it.

Information Systems are all over and we don't pay attention to them as we should do, instead, we are dealing with individual information focusing on our mission and forgetting where we are heading. That's why a strategic data-driven team, process, and tools aka *data strategy* need to be instrumented to take the organizations to the next level. In the business realm, those systems are sometimes called experience, structured tools (Structured problem solving), or even frameworks. Connecting all systems and doing mental effort daily on them is a full-time job that needs overseeing and needs high-end calibers to take over this hard mission and drive the organization towards its goals.

Discussing our main three goals to achieve through the technology and prove our case which are iterations, focus, and momentum. The frequency of high-quality iterations while focusing on findings and pivoting towards quicker winning tactics can accelerate the growth of any organization. The real-world case is that it isn't feasible to utilize data that much even if you did guarantee the fairness of data still you need to follow your gut. But the actual problem is that you don't know what you don't know and here comes the science one more time to say its word that is inescapable which is "You need to use informed decision-making approaches aka Data-Driven Decision-Making frameworks" which will need calibers to drive the process and achieving the needed quality and results.

Starting from the current state where there is a ton of waste all over in day-to-day business and the reason is that it is not feasible to be 100% data-driven, 100% documented, 100% aligned, etc. This is according to the limitations of the technologies being developed that leads to limitation in the applications of those technologies. That limitation is mainly because all dedicated resources get consumed in concrete features that eat up the investment and we end up with a very good solution but limited to the purpose it has been made for in the first place.

Utilizing the computer hardware we can operate on data in a different way where we keep the information in its syntactical form and operate on it without having to pass through the bottleneck back and forth.

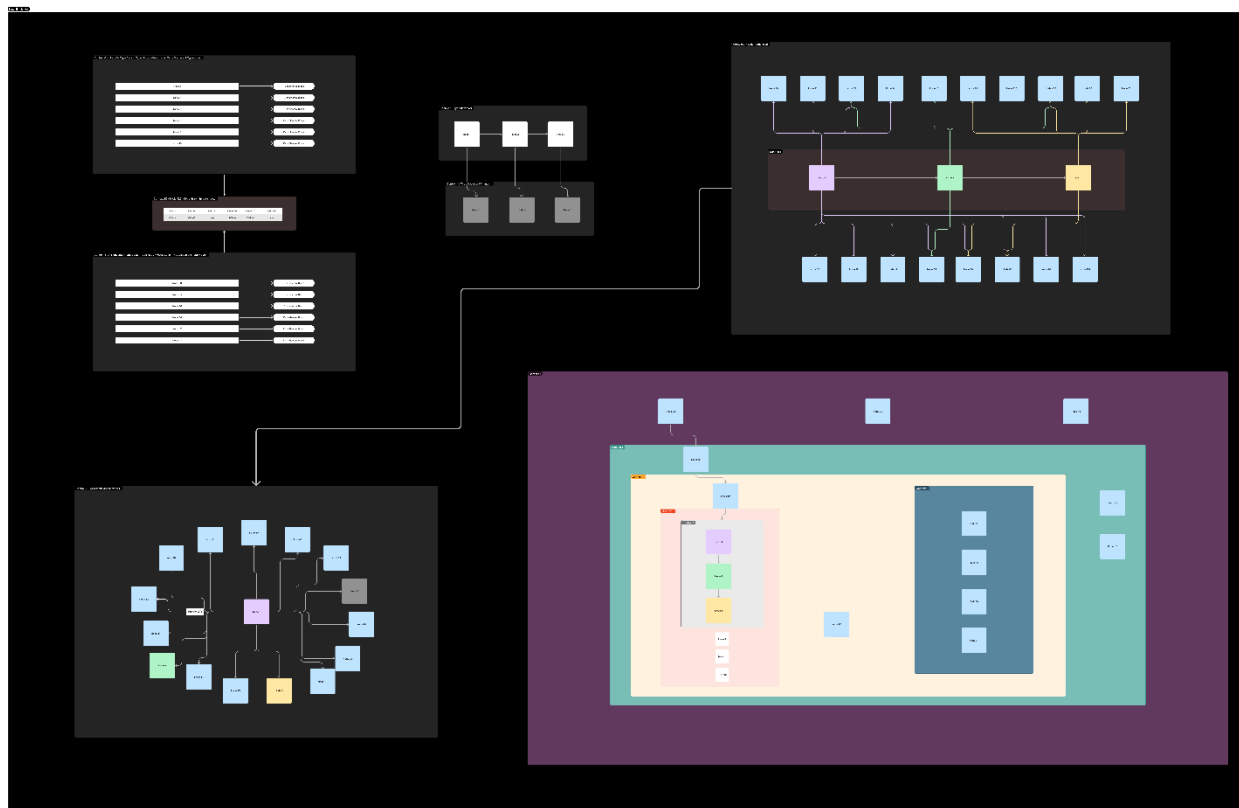


There are three types of bottlenecks. The first one is the software apps, tiers, architectures, compiling, and executing the operations on the information. The second one is the hardware bottleneck. The third is the Network bottleneck. The three bottlenecks are architectural bottlenecks where they suffocate the information, manipulate it, convert it to the machine code, move it through the bottleneck, and reverse the process back to

read the machine code and convert it back to its original form. And because of this bottleneck billions are spent monthly to solve for performance. Humanity is trapped in trying to optimize within the bottleneck it really starts when we ask the question, why we don't try to resolve the bottleneck or create sort of a worm whole where we can keep the application abstract and kill that waste for good?

If the problem is architectural, it is most likely the solution should be of the same nature as the problem. And I found the solution to that in the human mind where the most impressive sophisticated creation in the universe. I found that the human mind has architecture-less architecture. Neuroplasticity is to deform the whole architecture based on new inputs (information), attributes, and contexts around those. I found that information take any form and here I am telling how this technology can upgrade humanity and totally change the game.

After a demo of the sample data, the results were incredibly promising and in the near future, I can present the workable demo to the researchers to help me develop the missing parts of my master piece and to fill the gaps in my research with their specialized opinion each in his cross-functional department.



Here I demonstrate the architecture-less architecture based on sample data and how it is possible to build application based on our information system technology in that abstract way keeping the effort in the abstract state like powder the most.

One final sample to demonstrate the idea of the abstraction application is as follows:

The mathematical equation or any mathematical equation and the proof of it is of course close to 100% abstract, but why it is not 100% abstract? The answer is simple because it is not a text it is a mathematical notation at the end of the day and systems vary in efficiency depending on the type and the size of the problem you want to solve. For small linear problems you should use algebra “don’t use a cannon to kill a mosquito” and for complex graphical computational problems you can use linear algebra or even calculus for more advanced relative problems. Back to the topic researchers used to develop the theory on paper in

syntactical systemized notation like mathematical notation. However, the application of these computationally is always concrete. They even use just functions to run the mathematical operation and neglect the architecture because it is not their purpose to solve problem worlds, they only see what they are interested in. This makes the implementations, applications, and any form of executable thing digitally become concrete. Meaning that you have to repeat the effort to be able to change the application of the theory or at least make a huge effort. This can be a slightly small effort in the case of a small function but for systems like OpenGL, and CodeX for CG software it takes a lot to a slight change in the performance or to optimize the algorithm.

The change in the theory is abstract the most and the change in the application is highly concrete to an extent, and can we develop a technology that can resolve the bottleneck and keep everything abstract as it is in its informational form? With ISAAT we can surely say yes, we can.

This happens basically because the communication with the computer is in a huge bottleneck the humans need an interface with the computer that can help them communicate with it as if it they are physically wired to it, as if their minds are physically wired to the computers. The human then only can see and the clarity and precision are limitless. The cost spent on making the computer understand the human is tremendously huge. Progress is infinitesimal; thus humanity needs a core foundational solution to get out of that bottleneck and communicate with the computer in the most efficient powerful way which is through thoughts or meanings which is not possible today by any technology.

ISAAL literally upgrades human communication with himself and others.

Nowadays there are millions of apps in the market and dozens installed by the majority of consumers while users use very few of them. The waste is catastrophic and humanity is suffering because of the lack of vision and inability to force decision makers to follow science. There is new recent hope that investors are paying more attention to what they call investing by heart where human value is at the top of their priorities. But we need technology to speak to the majority and say our words utilizing the power of science to allow them to join the change wave and become more cooperative to eliminate the waste internally in their organizations and externally in the market.